

RESPONSE TO THE REVIEWERS

Manuscript Title: A Scalable Hadoop–Spark Framework for Large-Scale Carbon Emission Analytics and Forecasting

We sincerely thank the reviewers for their valuable comments and constructive suggestions. We have carefully revised the manuscript and addressed all concerns. The revisions have substantially improved the quality, clarity, and completeness of the paper. A detailed response to each comment is provided below.

Reviewer #1

Comment 1:

The manuscript lacks discussion of missing values and data quality issues in the OWID dataset.

Response:

A new subsection titled “Missing Data Analysis and Data Preprocessing” (Section 4.2) has been added. The revised manuscript now includes a detailed assessment of missing values across key variables, a summary table of missing-value statistics, and an explanation of the preprocessing strategy adopted. The rationale for using listwise deletion and feature selection has also been provided.

Comment 2:

The clustering analysis lacks interpretation and representative examples of countries within each cluster.

Response:

The clustering section has been expanded to include a detailed interpretation of the identified clusters. Representative countries are now reported for each cluster, and a cluster summary table has been added to improve interpretability. The discussion explains differences in emission behavior, economic scale, and population characteristics across clusters.

Comment 3:

Cluster labels and numbering require clarification.

Response:

Cluster numbering has been reviewed and corrected throughout the manuscript. Consistent cluster labels and numbering are now used in both the text and summary tables.

Comment 4:

Forecast uncertainty is not discussed.

Response:

A forecast uncertainty analysis has been incorporated into the LSTM forecasting section. Monte Carlo Dropout was implemented during inference using multiple stochastic forward passes. Mean forecasts and 95% confidence intervals are now reported and visualized in the forecasting figure. The discussion explains how uncertainty increases with forecast horizon and how confidence intervals improve the usefulness of forecasts for long-term planning.

Reviewer #2

Comment 1:

The manuscript claims scalability advantages but does not provide empirical scalability evidence.

Response:

A new subsection titled “Scalability Evaluation” (Section 4.8) has been added. A synthetic data augmentation experiment was conducted using datasets ranging from 50,191 to 500,000 records. Benchmarking was performed for aggregation, correlation analysis, and filtering operations in Apache Spark. Results demonstrate efficient scalability, with execution time increasing by only 1.29× despite a nearly tenfold increase in dataset size. A scalability benchmark table and scalability figure have been included to support these findings.

Comment 2:

The rationale for selecting Hadoop and Spark should be better justified.

Response:

The Discussion section has been expanded to explain the complementary strengths of Hadoop HDFS and Apache Spark. The revised manuscript now discusses distributed and fault-tolerant storage, parallel in-memory processing, scalability advantages, and benefits over traditional single-machine architectures.

Comment 3:

The abstract should better reflect the scalability contribution of the work.

Response:

The abstract has been revised to include the scalability benchmarking results and explicitly mention the validation of the proposed Hadoop–Spark architecture through scalability assessment. The revised abstract now highlights the efficient processing performance observed across datasets ranging from 50K to 500K records.

Programming and Computer Software (PCS) Requirements

Comment:

The manuscript should incorporate and discuss relevant literature published in Programming and Computer Software.

Response:

Three relevant PCS publications have been incorporated into the Related Works section and included in the reference list. The discussion now covers distributed computing frameworks, environmental analytics, and scalable real-time analytics systems, thereby strengthening the connection between the proposed work and prior PCS research.

Additional Revisions

- Added missing-value statistics table and supporting discussion.
- Added representative countries and interpretation for clustering results.

- Added forecast uncertainty estimation using Monte Carlo Dropout and 95% confidence intervals.
- Added scalability benchmarking experiments, table, figure, and discussion.
- Improved figure and table numbering consistency.
- Updated the abstract, discussion, and conclusion to reflect scalability findings.
- Performed language editing and formatting corrections throughout the manuscript.

We appreciate the reviewers' insightful comments, which have significantly improved the manuscript. We believe that all concerns have been satisfactorily addressed and respectfully submit the revised manuscript for consideration.